



# **BAAZ & UDAY JKBOSE**

**2026 CLASS 11<sup>TH</sup>**

**SETS**

**MATHS**

**LECTURE-06**

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# TOPICS TO BE COVERED

1

Operation on Sets

(Union of sets)

(Intersection of sets)

(Difference of sets)

2

Related Problems

5

# Operation on Sets



## \* Union of two sets:

- let  $A$  and  $B$  be any two sets, then their union is the collection of all the elements of  $A$  and  $B$ .

- Denoted by  $A \cup B$ .

- $A \cup B = \{x : x \in A \text{ or } x \in B\}$

$$* A = \{1, 2, 3, 4\} \text{ and } B = \{5, 6, 7\}$$

$$\Rightarrow A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$$

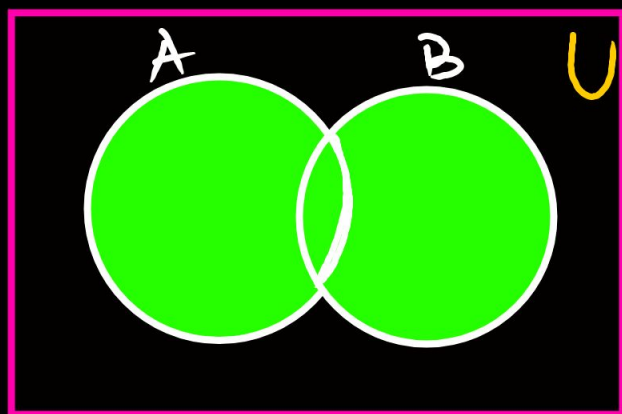
$$* A = \{1, 2, 4\} \text{ and } B = \{2, 4, 5\}$$

$$\Rightarrow A \cup B = \{1, 2, 4, 5\}$$

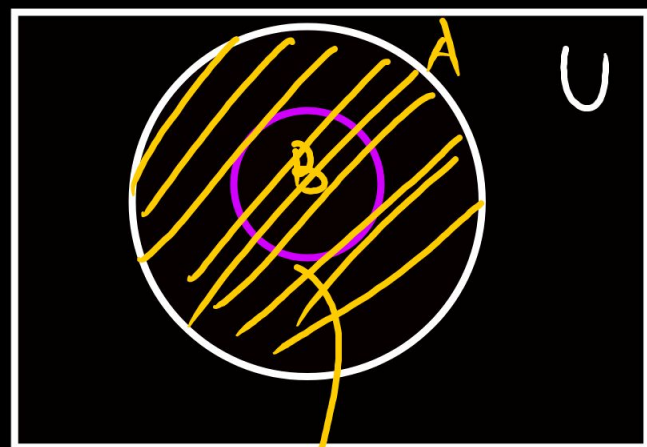
$$* A = \{1, 2, 5\} \text{ and } B = \{2, 5\}$$

$$\Rightarrow A \cup B = \{1, 2, 5\} = A$$

$$\Rightarrow A \cup B = A \Leftrightarrow B \subset A$$



$A \cup B$



$A \cup B = A$

## Intersection of two sets:

- let  $A$  and  $B$  be any two sets, then their intersection is the

collection of all the common elements of  $A$  and  $B$ .

- Denoted by  $A \cap B$ .

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

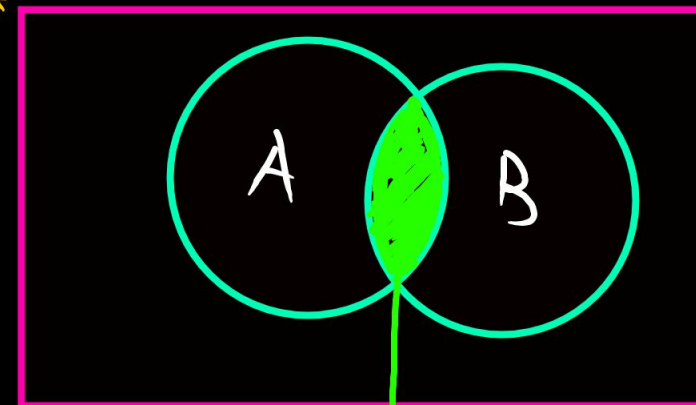
$$\begin{aligned} * A &= \{1, 2, 3, 4\} \text{ and } B = \{3, 4, 5\} \\ \Rightarrow A \cap B &= \{3, 4\} \end{aligned}$$

$$\begin{aligned} * A &= \{1, 2, 3, 4\} \\ \text{and } B &= \{3, 4\} \end{aligned}$$

$$\Rightarrow A \cap B = \{3, 4\} = B.$$

$$* A \cap B = B \Leftrightarrow B \subset A$$

\*



$A \cap B$

$$* A = \{1, 2, 3\} \text{ and } B = \{4, 5, 6\}$$

$$\Rightarrow A \cap B = \emptyset \text{ (Disjoint sets)}$$

• **Disjoint sets**: A and B are said to be disjoint sets if  $A \cap B = \emptyset$

$$* A = \{x : x \text{ is a vowel in English Alphabet}\}$$

$$* B = \{x : x \text{ is a Consonant in English Alphabet}\}$$

$$\Rightarrow A \cap B = \emptyset \text{ (Disjoint sets)}$$

$$* N \cup Z = Z ; N \cap Z = N.$$

$$* Z \cup Q = Q ; Z \cap Q = Z$$

$$* Q \cup R = R , Q \cap R = Q$$

**QUESTION**

If two sets  $A = \{1, 2, 3\}$  and  $B = \{1, 3, 5, 7\}$ , then  $A \cup B$  is equal to

- ☒ 1  $\{1, 2, 3, 7\}$
- ☒ 2  $\{1, 2, 3, 5, 7, 8\}$
- ☒ 3  $\{1, 2, 3, 5, 7\}$
- ☒ 4  $\{1, 2, 3, 4, 5, 6, 7\}$

**QUESTION**

If two sets  $A = \{1, 2, 3, 4\}$  and  $B = \{2, 4, 5\}$ , then  $A \cap B$  is equal to

☒ 1  $\{1, 3\}$

☒ 2  $\{2, 4\}$

☒ 3  $\{4, 3\}$

☒ 4  $\{1, 6\}$

## QUESTION

Given that  $A = \{1, 2, 3, 4, 5, 6\}$ ,  $B = \{7, 8, 9, 10, 11\}$ ,  $C = \{6, 8, 10, 12, 14\}$ . Which of the following pair of sets are Disjoint sets?

☒ 1 A and B ✓

$$A \cap B = \phi$$

$$B \cap C \neq \phi$$

$$A \cap C \neq \phi$$

☐ 2 B and C ✗

☐ 3 C and A ✗

☐ 4 None of the above

## QUESTION

Given that  $A = \{1, 2, 3\}$ ,  $B = \{3, 4\}$ ,  $C = \{4, 5, 6\}$ . Then,  $A \cup (B \cap C)$  is equal to

$$A = \{1, 2, 3\}$$

$$B \cap C = \{4\}$$

$$\Rightarrow A \cup (B \cap C) = \{1, 2, 3, 4\}$$

☒ 1  $\{3\}$

☒ 2  $\{1, 2, 3, 4\}$

☒ 3  $\{1, 2, 5, 6\}$

☒ 4  $\{1, 2, 3, 4, 5, 6\}$

# QUESTION

If  $X$  and  $Y$  are two sets such that  $n(X) = 17$ ,  $n(Y) = 23$  and  $n(X \cup Y) = 38$ , then  $n(X \cap Y)$

Is equal to

we know that

$$n(X \cup Y) = n(X) + n(Y) - n(X \cap Y)$$

$$\Rightarrow 38 = 17 + 23 - n(X \cap Y)$$

$$\Rightarrow 38 = 40 - n(X \cap Y)$$

$$\Rightarrow n(X \cap Y) = 40 - 38$$

$$\Rightarrow n(X \cap Y) = 2$$

Formula To be used:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

where  $A$  and  $B$  are finite sets.

\*\*\*

$$* n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

☒ 1 4

☒ 2 2

☒ 3 6

☒ 4 3

# QUESTION (NCERT EXEMPLAR):

If  $A$  and  $B$  are two sets, then  $A \cap (A \cup B)$  equals

1

$A$

~~2~~

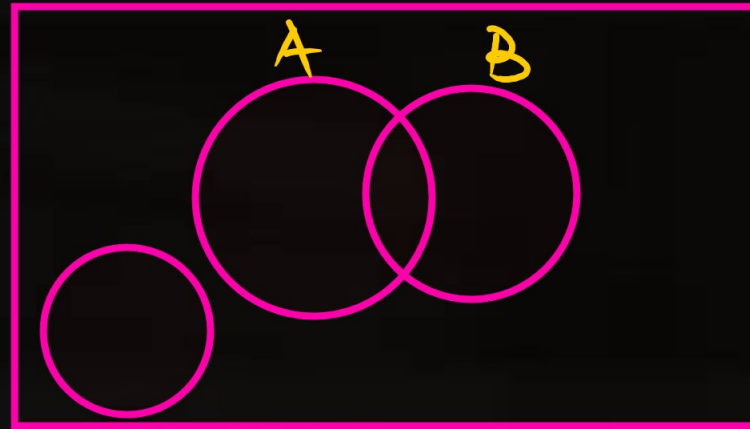
$B$

~~3~~

$\phi$

~~4~~

$A \cap B$



$$A \cap (A \cup B) = A$$

$$B \cap (A \cup B) = B.$$

$$B = \{2, 4\} \quad A = \{1, 2, 4\}$$

$$A \cap B = B \Leftrightarrow B \subset A.$$

$$A \cup B = A \Leftrightarrow B \subset A.$$

$$* A = \{1, 2, 3\}, B = \{3, 4, 5\}$$

$$C = \{4, 5, 6\}, D = \{6, 7\}$$

$$\Rightarrow A \cup B \cup C = \{1, 2, 3, 4, 5, 6\}$$

# Properties of Union and Intersection

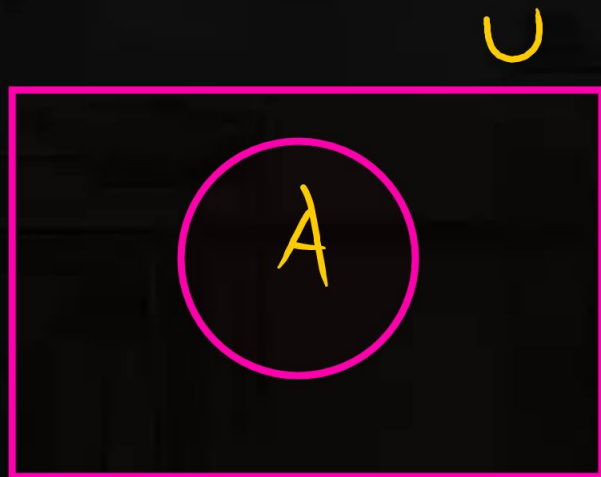
(i)  $A \cup B = B \cup A$  (Commutative law)

(ii)  $(A \cup B) \cup C = A \cup (B \cup C)$  (Associative law)

(iii)  $A \cup \phi = A$  (Identity law)

(iv)  $A \cup A = A$

(v)  $U \cup A = U$



(i)  $A \cap B = B \cap A$  (Commutative law)

(ii)  $(A \cap B) \cap C = A \cap (B \cap C)$  (Associative law)

(iii)  $\phi \cap A = \phi, U \cap A = A$

(iv)  $A \cap A = A$

(v)  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

(Distributive law)

\*  $\cap$  Distributes over  $\cup$ .

## QUESTION (NCERT EXEMPLAR):

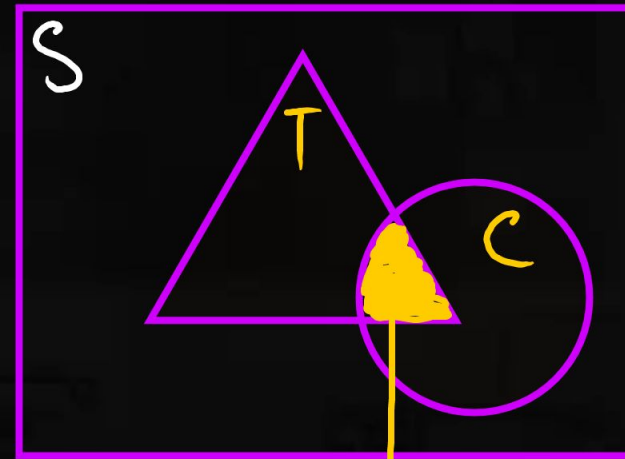
Let  $S$  = set of points inside the square,  $T$  = the set of points inside the triangle and  $C$  = the set of points inside the circle. If the triangle and circle intersect each other and are contained in a square. Then.

~~1~~  $S \cap T \cap C = \phi$

~~2~~  $S \cup T \cup C = C$

☒ 3  $S \cup T \cup C = S$

~~4~~  $S \cup T = S \cap C$



$S \cap T \cap C \neq \phi$

# QUESTION (NCERT EXEMPLAR): *(Home work)*

Let  $F_1$  be the set of parallelogram,  $F_2$  the set of rectangles,  $F_3$  the set of rhombuses,  $F_4$  the set of squares and  $F_5$  the set of trapezium in a plane. Then  $F_1$  may be equal to

**1**  $F_2 \cap F_3$

**2**  $F_3 \cap F_4$

**3**  $F_2 \cup F_5$

**4**  $F_2 \cup F_3 \cup F_4 \cup F_1$

# QUESTION(JEE MAINS-2021)

In a School, there are three types of games to be played. Some of the students play two types of games, but none play all the three games. Which Venn diagram can justify the above Statement

~~1~~

P

~~2~~

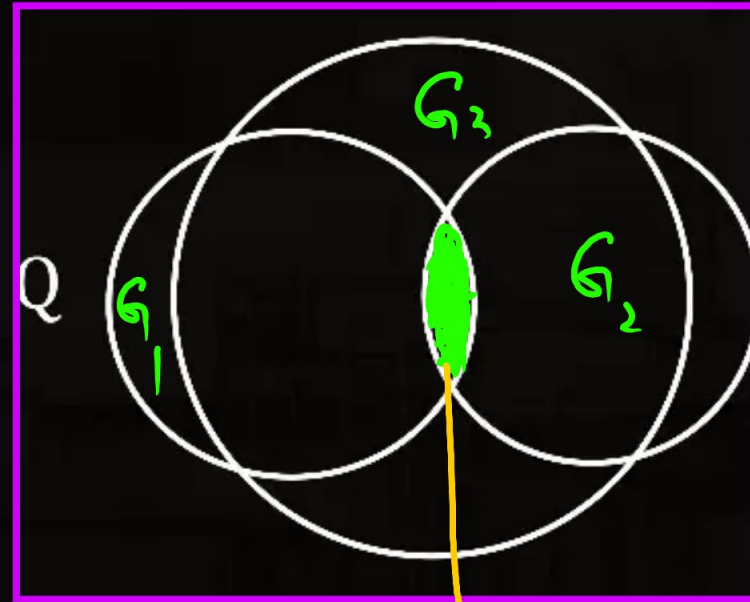
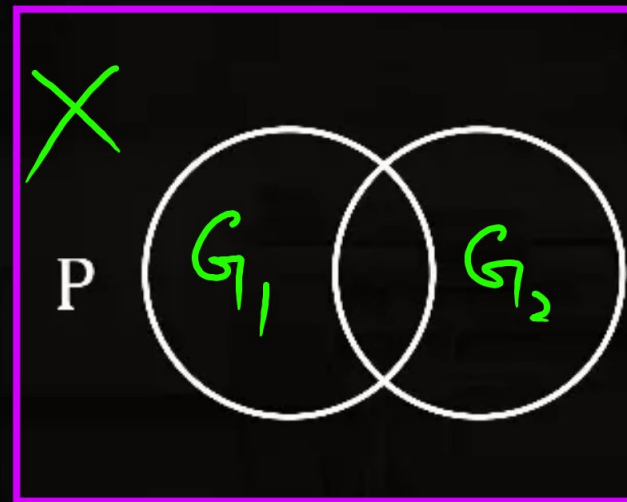
Q

~~3~~

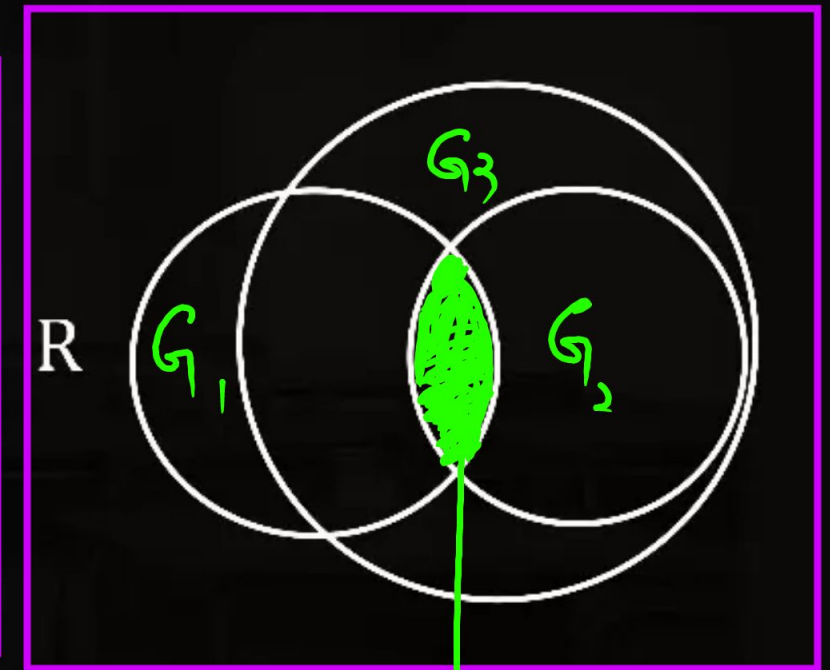
R

4

None of these



$$G_1 \cap G_2 \cap G_3 \neq \phi$$



$$G_1 \cap G_2 \cap G_3 \neq \phi$$

## Difference of sets

let  $A$  and  $B$  be any two sets, then

their difference,  $A - B$  is the collection of all the elements of  $A$  which are not present in  $B$ .

$$\text{i.e. } A - B = \{x : x \in A \text{ and } x \notin B\}$$

$$* B - A = \{x : x \in B \text{ and } x \notin A\}$$

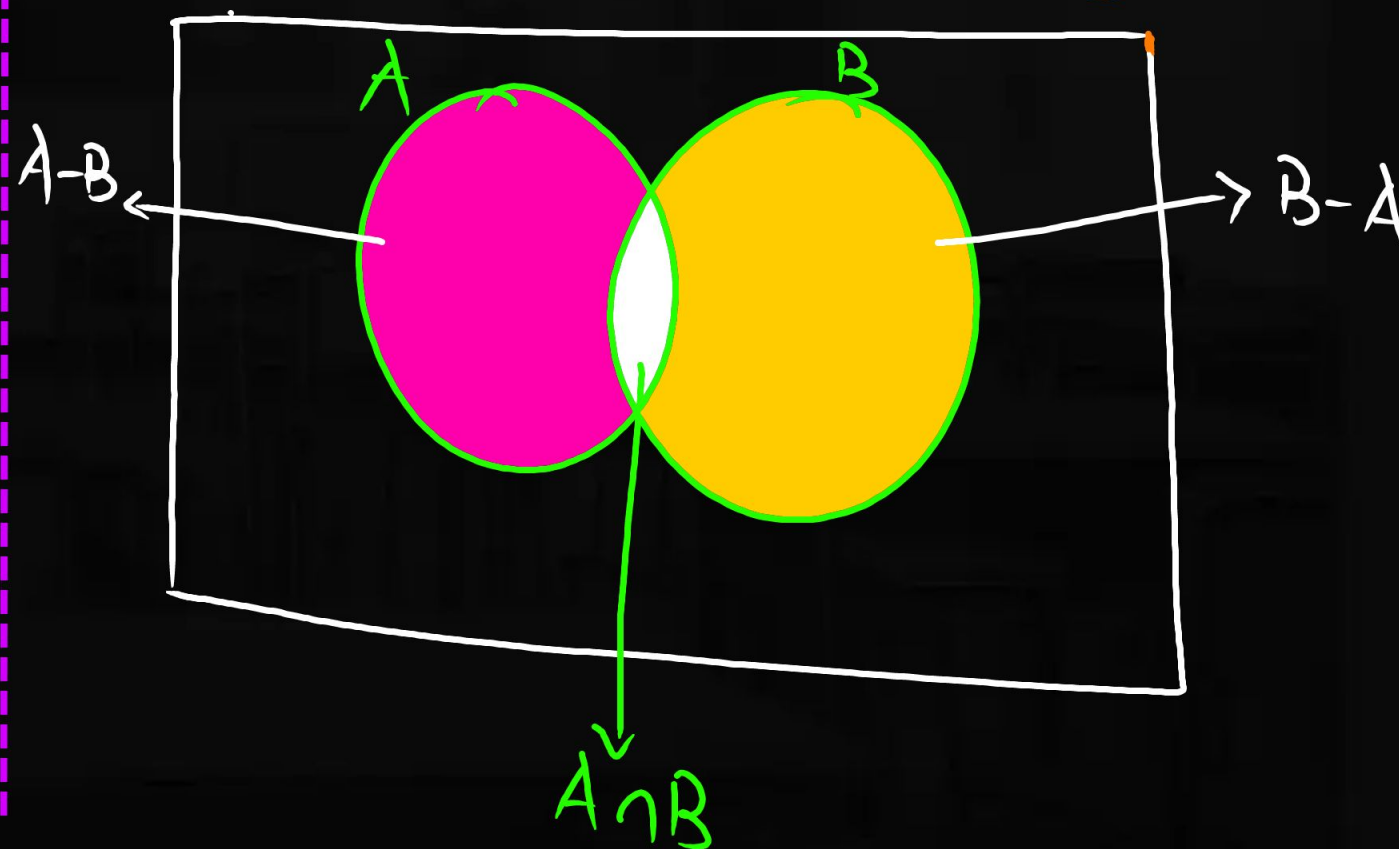
$$* A = \{1, 2, 3, 4\} \text{ and } B = \{3, 4, 6, 7\}$$

$$\Rightarrow A - B = \{1, 2\}$$

$$\text{and } B - A = \{6, 7\}$$

$$* (A - B) \cup (A \cap B) = A$$

$$* (B - A) \cup (A \cap B) = B$$



**Question**

If  $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$  and  $B = \{1, 4, 7, 8, 9\}$ , then  $A - B$  is equal to

**1**  $\{1, 2, 3, 4, 5\}$  ✗  $A \cap B = \{1, 4, 7, 8, 9\}$

**2**  $\{2, 3, 6, 4, 5\}$  ✗

**3**  $\{2, 3, 5, 6\}$  ✓

**4** None of the above ✗

## Question

If  $R$  is the set of Real numbers and  $Q$  is the set of Rational numbers, then  $R-Q$  is

~~1~~

The set of Integers

~~2~~

The set of Non negative Real numbers

~~3~~

The set of Irrational Numbers

~~4~~

The set of negative Real numbers

## Question

If  $A = [-2, 4]$  and  $B = [-1, 3]$ , then  $A - B$  is equal to:

**1**  $(-2, -1) \cup (3, 4]$

(Homework)

**2**  $[-2, -1] \cup [3, 4]$

**3**  $[-2, -1) \cup (3, 4]$

**4**  $[-2, -1] \cup (3, 4]$

**Thank**  
**You**